

COMPREHENSIVE SPACE & UTILIZATION EVALUATION

Classroom Building | Chemistry Building | Olson Hall

University of the Pacific | Stockton, California

Final Evaluation Report

LEAD 215: Evaluation for Organizational Effectiveness

Evaluation Team

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Spring 2026

Benerd College | Master of Arts in Organizational Leadership

Table of Contents

Executive Summary

Part 1: Purpose, Goals, and Context

1.1 Evaluation Purpose

1.2 Evaluation Background

1.3 Context

1.4 Stakeholders

1.5 Key Questions

1.6 Logic Model

Part 2: Program Evaluation Plan

2.1 Type of Evaluation

2.2 Evaluation Model

2.3 Data Collection Plan

2.4 Limitations

2.5 Project Management Plan

Part 3: Findings and Action Planning

3.1 Data Summaries and Visualization

3.2 Communication and Collaboration Approach

3.3 Stakeholder-Developed Action Plan

3.4 Evaluator Recommendations

Appendix A: Data Collection Tools

Appendix B: Floor Plans

Executive Summary

This report presents the findings of a comprehensive space and utilization evaluation conducted on three high-traffic instructional assets at the University of the Pacific's Stockton campus: the Classroom Building, the Chemistry Building, and Olson Hall. The evaluation was commissioned by President Callahan and the Senior Vice President of Facilities in direct support of the university's exploration of establishing a new medical school on the Stockton campus, a strategic initiative driven by the severe physician shortage across California's Central Valley region.

Our evaluation team, consisting of Sam B., Sam C., and Veronica H., conducted this study over the Spring 2026 semester as the culminating project for LEAD 215: Evaluation for Organizational Effectiveness. The project employed a Utilization-Focused Evaluation (UFE) model within a developmental evaluation framework, recognizing that the university's decision-making environment is complex and rapidly evolving. Data collection utilized a mixed-methods approach combining quantitative archival data extraction from the university's EMS and Banner scheduling systems with qualitative key informant interviews of department administrators and facilities staff.

Key Findings. Analysis of Spring 2026 scheduling data across the three buildings revealed the following:

The **Classroom Building** is the campus's heaviest instructional asset, carrying 77 scheduled sections across 14 rooms for approximately 229.63 total weekly scheduled hours, with an average seat-fill rate of 0.63. This moderate fill rate suggests meaningful consolidation potential through scheduling optimization. The **Chemistry Building** carries 34 sections totaling 96.33 weekly hours with the highest average seat-fill of 0.76. Because all Chemistry instruction is laboratory-based, this building is comparatively harder to relocate without specialized swing space. **Olson Hall** is the smallest of the three, with 29 sections across 5 rooms and 74.33 weekly hours at an average seat-fill of 0.66. A few Olson rooms are heavily utilized and should be preserved as priority swing space.

A small number of "workhorse" rooms carry a disproportionate share of the instructional load: Classroom 205 (~31 weekly hours), Classroom 203 (~29.25), Olson 120 (~25.75), and Classroom 104 (~25.75). Replacement plans must ensure equivalents for these rooms are prioritized. Simultaneously, several rooms showed consistently low utilization, including Olson 103 (avg seat-fill 0.29), Olson 105 (0.38), and Classroom 126 (0.44), representing immediate candidates for consolidation or repurposing.

Using our composite Efficiency Score formula ($100 * [0.65 * \text{Utilization} + 0.35 * \text{Intensity}]$), the top-performing rooms were Olson 120 (84.4), Classroom 104 (80.8), and Classroom 238 (79.7) among lecture spaces, and Chemistry 101 (77.0) and Chemistry 203 (73.0) among lab spaces.

Recommendations. Based on the evaluation findings and collaborative stakeholder sessions, we recommend three tiers of action: (1) immediate scheduling optimizations to improve seat-fill without capital investment, such as right-sizing repeated low-fill rooms and redistributing peak-period loads; (2) medium-term swing space strategies identifying which rooms can absorb displaced instruction if a building goes offline for construction; and (3) long-term capital planning guidance for the Board of Regents regarding the medical school site, informed by the provost's feasibility study confirming the university's institutional readiness and the projected \$150M capital requirement. These findings were

presented to President Callahan and university leadership and contributed to the ongoing go/no-go deliberations for the medical school initiative.

Part 1: Purpose, Goals, and Context

1.1 Evaluation Purpose

The aim of this evaluation is to transition the university's space management model from an anecdotal, decentralized perspective to a data-driven analytical framework. The results of this analysis, specifically the review and quantification of the "efficiency gap" between planned instructional hours and actual seat occupancy, serve as a strategic baseline for President Callahan and the Board of Regents to make high-level decisions about capital planning, possible building changes, and the viability of the proposed medical school campus expansion.

1.2 Evaluation Background

The university currently relies on legacy scheduling systems (Banner, EMS) to manage its primary instructional and administrative assets, including the Classroom Building, Chemistry Building, and Olson Hall. There is increasing concern that these facilities are being suboptimally utilized across both revenue-generating instructional capacity and administrative overhead. This assessment was needed to ensure factual accuracy of current records, pinpoint what "headroom" exists for possible consolidation, and furnish quantitative evidence for future maintenance or renovation budgets in advance of the Board of Regents' April session.

The initiative was catalyzed by President Callahan's proposal to establish a medical school on the Stockton campus. A global consulting firm hired by the university completed a comprehensive feasibility study in Fall 2025, confirming that Pacific has "the institutional maturity and infrastructure to launch a medical school." The feasibility study projected an inaugural cohort of 60 students by Fall 2030, growing to 400 students by Fall 2037, with an estimated \$150M in one-time capital costs. However, the consultants noted that while the Stockton campus has "ample space and infrastructure," targeted renovations and new construction are still required. Our space utilization evaluation provides the granular, building-level data necessary to determine which spaces can be repurposed, consolidated, or relocated to accommodate this expansion.

1.3 Context

This evaluation takes place at a pivotal moment for the University of the Pacific. The university is considering significant structural changes, including the potential decommissioning or complete replacement of aging facilities like the Classroom Building, to support the medical school initiative. These decisions sit at the crossroads of Academic Affairs (scheduling and pedagogy), Finance (capital expenditure), and Physical Plant (maintenance and operations), requiring cross-departmental agreement on how physical space aligns with the institution's educational mission.

The broader strategic context includes California's severe physician shortage. Thirteen California counties, including all eight Central Valley counties south of Sacramento, fall below the minimum recommended range of 60 primary care physicians per 100,000 residents. The feasibility study demonstrated that physicians tend to practice near where they attended medical school, with California showing an 82% retention rate for graduates who completed both medical school and

post-graduate education in-state. This presents a compelling case for Pacific to address a regional need while strengthening its institutional brand as the only university among its peers with schools of medicine, pharmacy, dentistry, and health sciences.

1.4 Stakeholders

Program End-Users

Students: Students are the primary users of the instructional spaces under evaluation. Their educational experience is directly impacted by room quality, seat availability, and the proximity of lab and lecture facilities. Any consolidation or relocation plan must preserve instructional quality.

Faculty and Staff: Occupants of the administrative offices and researchers utilizing the labs rely on these spaces for daily professional functions and instructional delivery. Faculty in Chemistry, Physics, and Mathematics have specialized space requirements that cannot be easily substituted.

Client / Partner

President Callahan, the Chief Financial Officer, College of the Pacific, and the Senior Vice President of Facilities: The executive sponsors who commissioned this evaluation to understand the "why" behind future capital requests and to prepare for the Board of Regents presentation. The VP of Facilities specifically tasked our team with producing the space utilization data and floor plan analysis for the April board meeting.

Other Stakeholders

Board of Regents: Holds the ultimate fiduciary responsibility for approving major renovations or new construction; they require the evaluation's data to mitigate financial risk in the context of the projected \$150M medical school investment.

The Registrar's Office: Responsible for maintaining EMS and Banner data; they have a vested interest in data accuracy and improved scheduling efficiency.

Department Chairs and Administrators (Math, Physics, Chemistry): Oversee the "live rosters" for their departments and have a vested interest in ensuring their faculty and specialized lab needs are not marginalized during any consolidation planning.

Physical Plant / Facilities Management: Responsible for long-term maintenance of these assets; they will use the findings to prioritize maintenance scheduling and capital improvement requests.

1.5 Key Questions

The following questions guided the evaluation:

1. To what extent does the actual seat-fill ratio (utilization) differ from the scheduled capacity across the three target buildings?
2. How efficient is the use of specialized "Lab" spaces in the Chemistry Building compared to general "Lecture" spaces in the Classroom Building and Olson Hall?
3. What is the current "Efficiency Gap" (the difference between scheduled hours and actual use) and where does the greatest potential for space consolidation lie?
4. How accurate are the current systems of record (EMS/Banner) relative to real daily levels of occupancy and administrative footprint for these buildings?

1.6 Logic Model of the Evaluand

The evaluand is the university's current space management program. The logic model below represents how the program is theorized to function, providing a baseline for determining whether it works in practice as intended. The primary metric used is the Efficiency Score: $100 * [0.65 * \text{Utilization} + 0.35 * \text{Intensity}]$.

Component	Description
Context	Aging campus infrastructure; manual/anecdotal space allocation; upcoming April Board of Regents deadline for relocation/renovation decisions.
Assumptions	Current scheduling data is partially outdated; lab space is more expensive to maintain than lecture space; higher seat-fill indicates better resource stewardship.
Resources / Inputs	Staff: Registrar, Dept. Admins, Dept. Chairs, Physical Plant. Systems: Banner, EMS. Physical: 3 buildings (Classroom, Chemistry, Olson Hall). Data: enrollment rosters, room capacities, floor plans, key informant interviews.
Activities	Scheduling of courses; assignment of faculty offices; tracking of lab usage; maintenance of physical room features; manual entry of section data into Banner.
Outputs	Weekly room-hour reports; department-specific schedules; office occupancy lists; room efficiency scores; color-coded floor plan visualizations.
Outcomes (Short-term)	Identification of underutilized rooms; correction of outdated room data; improved visibility into peak occupancy times.
Impact (Long-term)	Data-driven capital planning; optimized instructional vs. administrative footprint; successful medical school development/relocation based on validated space needs.

Part 2: Program Evaluation Plan

2.1 Type of Evaluation

We conducted a **Developmental Evaluation**. This type is most appropriate for initiatives operating in complex, uncertain, and rapidly changing environments. The university is in the early stages of deciding whether and how to expand its medical school, potentially by moving or remodeling major campus areas including the Classroom Building, Chemistry Building, and Olson Hall. Because no final "go/no-go" decision had been reached at the time of this evaluation, a developmental approach enabled our team to test assumptions about current space utilization in real time, refine strategies as new information from EMS and Banner was discovered, and provide ongoing, practical feedback to leadership. We evaluated the current state of space management to guide the design of the future campus footprint.

2.2 Evaluation Model

The project was predominantly guided by the **Utilization-Focused Evaluation (UFE)** model (Patton, 2008). The UFE model is founded on the principle that an evaluation should be judged by its actual utility to its intended users. Our principal clients, President Callahan, the VP of Facilities, and the Board of Regents, commissioned this study under a tight deadline (the April Board meeting) to address specific questions around capital planning and the "efficiency gap."

By applying UFE principles, we designed our data points and measures, specifically the composite Efficiency Score ($100 * [0.65 * \text{Utilization} + 0.35 * \text{Intensity}]$), to deliver actionable intelligence for high-stakes financial and structural decisions regarding the medical school proposal. This ensured we did not gather information for information's sake, but instead produced findings directly aligned with the Board's requirement for quantitative evidence before approving renovation or relocation budgets.

2.3 Data Collection Plan

To answer the key questions established in Part 1, we utilized a mixed-methods approach combining real archival data with primary qualitative data.

Tool 1: Archival Data Extraction (Quantitative)

Description: We extracted and cross-referenced space utilization data from the university's legacy systems of record (EMS and Banner). A standardized extraction protocol was used to pull raw room-hour scheduling, enrollment caps, and actual seat-fill data.

Approach: Quantitative analysis. We calculated efficiency scores for all rooms across the three target buildings to identify the "Efficiency Gap."

Population/Sampling: 100% census of all scheduled instructional spaces (Lecture and Lab) within the Classroom Building, Chemistry Building, and Olson Hall for the Spring 2026 semester.

Solicitation: Formal data request submitted via email to the Registrar's Office and the VP of Facilities (see Appendix A).

Tool 2: Key Informant Interviews (Qualitative)

Description: Semi-structured, 20-minute interviews with department administrators and facilities staff.

Approach: Qualitative verification. Our team noted discrepancies in the systems of record (e.g., unlisted faculty offices and intentionally blank basement floor plans). Interviews contextualized the quantitative data, revealing "off-the-books" usage of lab and office spaces.

Population/Sampling: Purposive sampling of 4-5 key informants: Department Admins for Math, Physics, and Chemistry, plus a representative from Physical Plant.

Solicitation: Direct email invitations sent from the evaluation team, referencing the VP of Facilities' sponsorship of the project (see Appendix A).

Data Collection Matrix

Key Question	Data Collection Tool	Data Points	Source
Q1: Seat-fill ratio vs. scheduled capacity	Archival Data Extraction	Room capacity, enrollment counts, weekly hours, seat-fill ratios	EMS / Banner / Registrar
Q2: Lab vs. lecture efficiency	Archival Data Extraction	Space type classification, efficiency scores by type	EMS / Academic Affairs
Q3: Efficiency gap and consolidation potential	Archival Data + Interviews	Scheduled hours vs. actual use; "off-books" usage	EMS / Banner / Dept. Admins
Q4: System accuracy	Key Informant Interviews	Discrepancies between records and reality; office assignments	Dept. Admins / Physical Plant

2.4 Limitations

System Inaccuracies: The archival data from EMS and Banner is known to be partially outdated or manually entered with errors. *Mitigation:* We used the second tool (key informant interviews) to cross-verify the systemic data with the people who actually manage the live rosters.

Narrow Scope / Time Constraints: Due to the April Board of Regents deadline, we evaluated only three buildings rather than the entire campus. *Mitigation:* The Classroom Building, Chemistry Building, and Olson Hall represent the highest-traffic instructional assets; evaluating them provides a sufficient, representative baseline to inform the medical school feasibility discussion without exceeding the project timeline.

Self-Report Bias: Interview subjects may overstate their need for space to avoid losing departmental real estate. *Mitigation:* Triangulating interview responses with the hard quantitative utilization data allowed us to objectively verify these claims.

2.5 Project Management Plan

Task Responsibilities

Veronica H.: Lead Liaison for Registrar's Office. Responsible for securing EMS/Banner data access, verifying room/class data, and identifying discrepancies in the system.

Samuel C.: Lead for Stakeholder Engagement. Responsible for conducting key informant interviews with department admins (Math, Physics, Chemistry) and managing communications with the client.

Sam B.: Lead Data Analyst. Responsible for processing the raw scheduling data, calculating efficiency scores, and developing the color-coded floor plan visualizations used in the final presentation to stakeholders.

Timeline (Spring 2026)

Phase	Dates	Activities
Phase 1	Jan 15 - Jan 31	Finalize evaluation plan; secure instructor approval for tools; send data requests to Registrar.
Phase 2	Feb 1 - Feb 28	Extract archival data; run efficiency score formulas; conduct key informant interviews.
Phase 3	Mar 1 - Mar 15	Complete cross-reference analysis; develop color-coded floor plans and data visualizations.
Phase 4	Mar 16 - Mar 31	Finalize executive report; present findings to President Callahan and Board of Regents.

Budget and Resources

Financial Budget: \$0 external costs. All labor was provided in-kind by the student evaluation team.

Resources Used: Access to EMS/Banner extraction reports; university floor plans; Zoom (for virtual stakeholder interviews); spreadsheet and data analysis tools.

Part 3: Findings and Action Planning

3.1 Data Summaries and Visualization

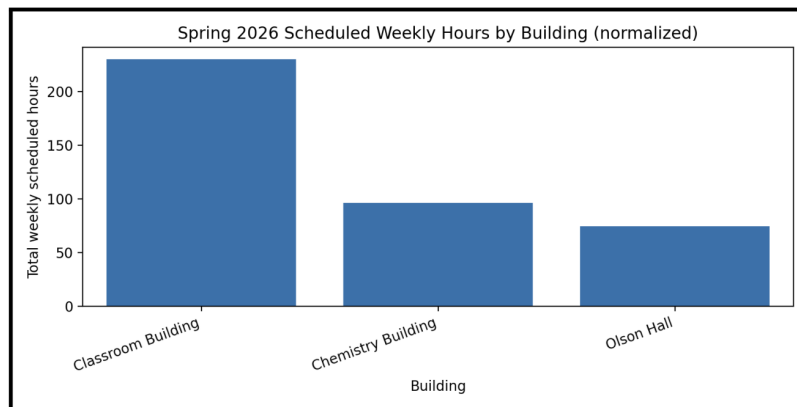
Finding 1: Building-Level Load and Relative Replaceability

Analysis of Spring 2026 scheduling data revealed significant variation in instructional load across the three buildings:

Building	Sections	Rooms	Weekly Hours	Avg Seat-Fill
Classroom Building	77	14	229.63	0.63
Chemistry Building	34	8	96.33	0.76
Olson Hall	29	5	74.33	0.66

The Classroom Building carries the largest single instructional burden to replace. Chemistry is smaller in total hours but runs laboratory instruction with higher seat utilization (less slack), making it comparatively harder to relocate without specialized swing space.

Total weekly scheduled hours by building



Classroom Building is the largest single instructional burden to replace; Chemistry is smaller in total hours but runs laboratory instruction with higher seat utilization (less slack). Olson is smaller but contains some high-use rooms.

Insight 2 — Seat-fill and consolidation potential

Seat-fill boxplots by building

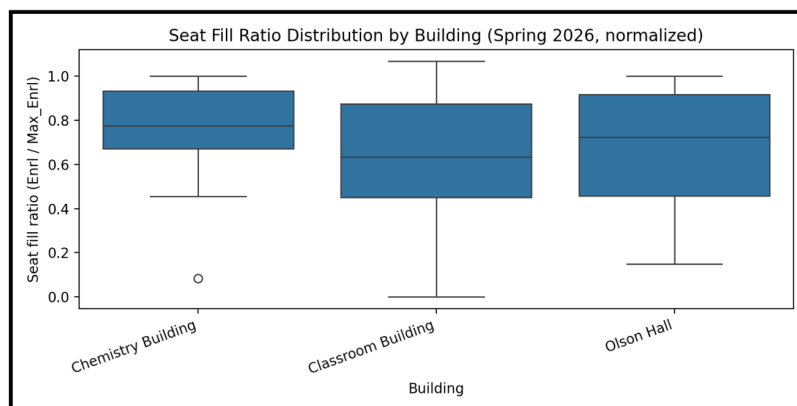


Figure 1: Total Weekly Scheduled Hours by Building and Seat-Fill Boxplots

Finding 2: Seat-Fill and Consolidation Potential

Classroom Building and Olson Hall show more low-fill sections, indicating greater potential for right-sizing or consolidation. Chemistry's higher fill suggests fewer simple consolidation opportunities for lab sections without changing schedules or capacities.

Finding 3: Workhorse Rooms

A relatively small number of rooms account for a disproportionate share of scheduled time. Replacement plans must ensure equivalents for these rooms (capacity, layout, and technology) are prioritized.

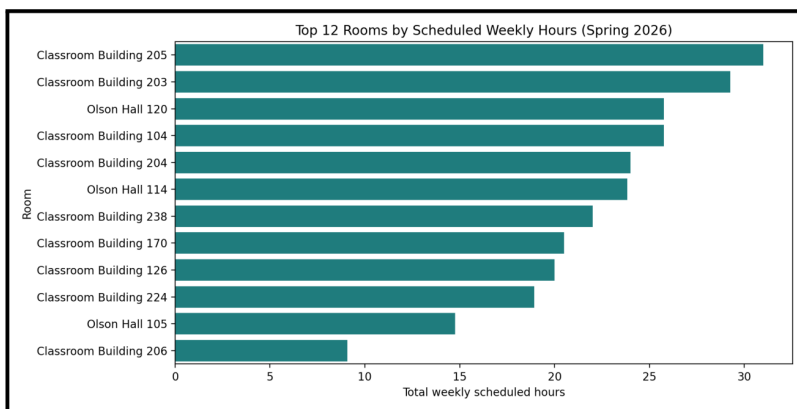
Room	Weekly Hours	Avg Seat-Fill	Efficiency Score
Olson 120	25.75	0.851	84.4
Classroom 104	25.75	0.796	80.8
Classroom 238	22.0	0.844	79.7
Olson 114	23.83	0.793	78.4
Classroom 203	29.25	0.689	77.8
Classroom 205	31.0	0.584	72.9
Chemistry 101	22.67	0.647	77.0
Chemistry 203	14.17	0.854	73.0

Classroom Building and Olson Hall show more low-fill sections (more potential to right-size or consolidate). Chemistry's higher fill suggests fewer simple consolidation opportunities for lab sections without changing schedules or capacities.

Insight 3 — Small set of “workhorse” rooms carry a lot of hours

- Data: top rooms by scheduled weekly hours (leading rooms across the three buildings)
 - Example top rooms (selected):
 - Classroom 205 (~31 weekly hours)
 - Classroom 203 (~29.25)
 - Olson 120 (~25.75)
 - Classroom 104 (~25.75)
 - Classroom 204 (~24.0)
 - Olson 114 (~23.83)
 - Classroom 238 (~22.0)

Top rooms bar chart



A relatively small number of rooms account for a disproportionate share of scheduled time. Replacement plans should ensure equivalents for these rooms (capacity, layout and technology) are prioritized.

Insight 4 — When instruction occurs (peak-period exposure)

- Data / Visuals:
 - **day-of-week × hour heatmaps** for buildings show peak daytime weekday loads and the exact blocks where instruction concentrates.

Figure 2: Top Rooms by Scheduled Weekly Hours

Finding 4: Peak-Period Exposure

Day-of-week by hour heatmaps reveal that peak instruction concentrates during midday weekday blocks (particularly Monday through Thursday, 10am-2pm). Any swing space plan must absorb classes during these same peak blocks, not just match total weekly hours.

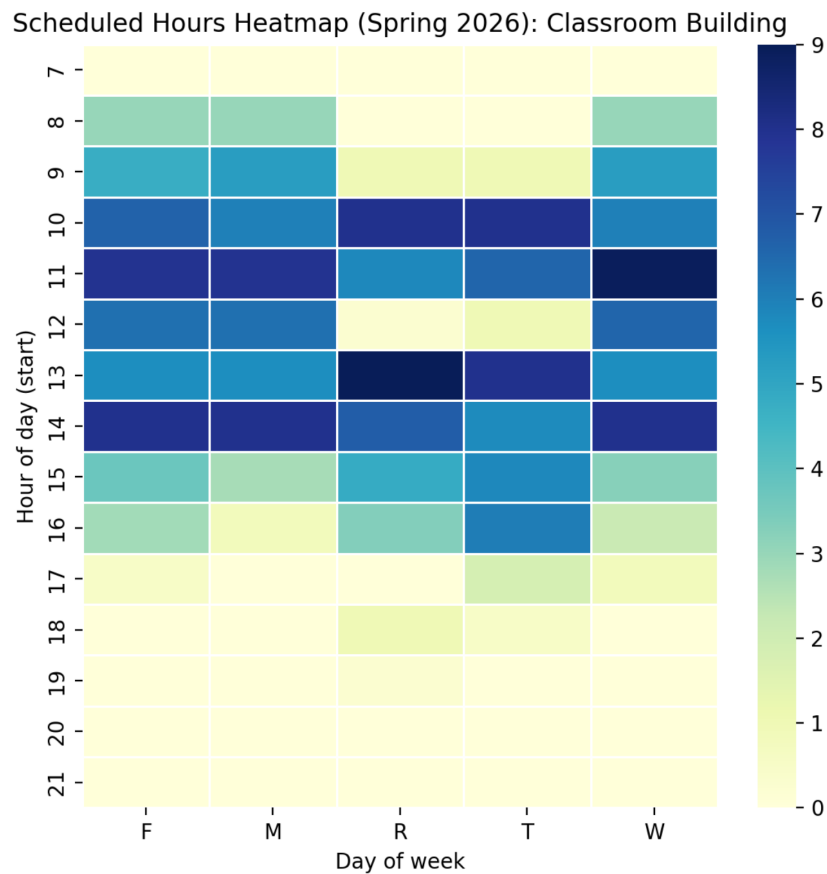


Figure 3: Classroom Building - Scheduling Heatmap

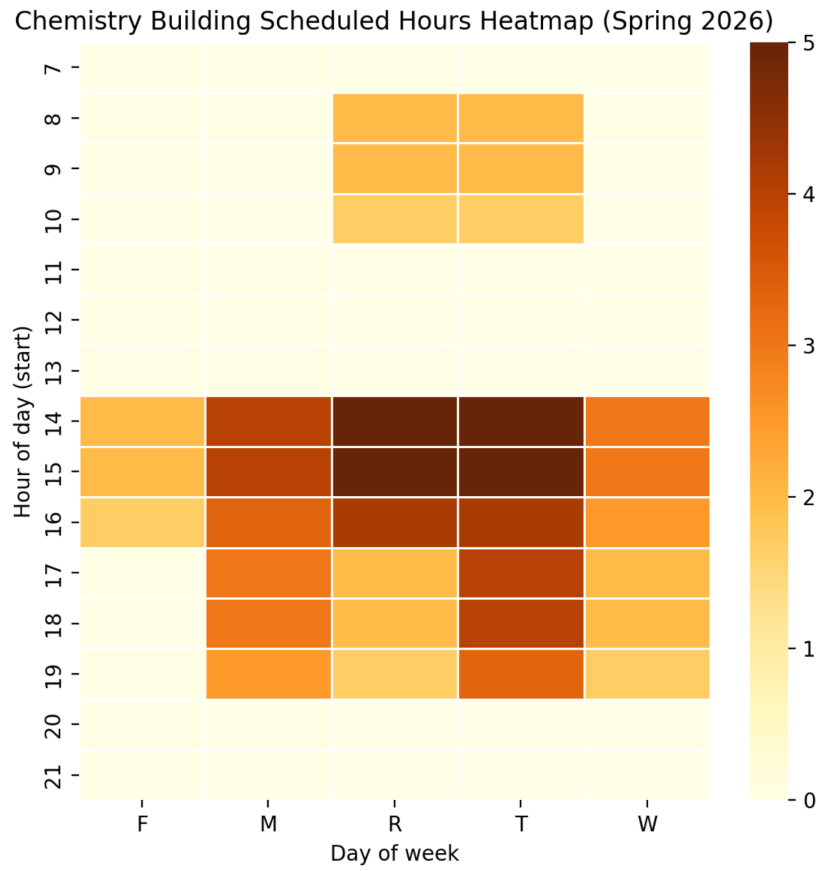
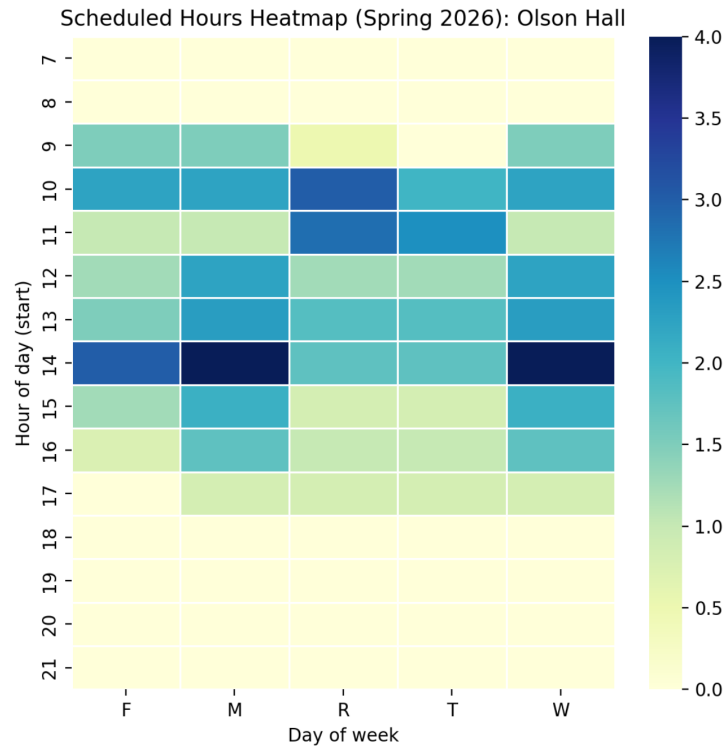


Figure 4: Chemistry Building - Scheduling Heatmap



Peak blocks (midday, weekdays) are critical. Any swing space plan must absorb classes during these same blocks, not just match total weekly hours.

Figure 5: Olson Hall - Scheduling Heatmap

Finding 5: Underutilized Rooms

Several rooms showed consistently low utilization and represent immediate candidates for consolidation or repurposing:

Room	Meetings	Avg Seat-Fill	Weekly Hours	Efficiency Score
Olson 100	1	0.15	1.25	11.2
Classroom 201	1	0.275	7.97	26.9
Olson 103	3	0.308	8.75	29.9
Classroom 232	1	0.433	4.0	32.7

Room	Meetings	Avg Seat-Fill	Weekly Hours	Efficiency Score
Classroom 115	1	0.50	2.83	35.7
Classroom 235	3	0.444	8.5	38.5
Olson 105	5	0.399	14.75	42.6

Right-sizing can start with repeated low-fill rooms and single/low-hour rooms to create immediate slack for peak-period absorption during any construction or transition period.

Finding 6: Efficiency Score Distribution

The composite Efficiency Score (0-100) combines seat-fill utilization (weighted 65%) and scheduled intensity (weighted 35%), normalized within space type so labs are compared to labs and lecture rooms to lecture rooms. The distribution reveals a wide performance range, with lab spaces clustering in the 54-77 range and lecture spaces spanning from 11 to 84.

Room	Type	Sections	Weekly Hrs	Seat-Fill	Score
Olson 120	Lecture	7	25.75	0.851	84.4
Classroom 104	Lecture	7	25.75	0.796	80.8
Classroom 238	Lecture	6	22.0	0.844	79.7
Olson 114	Lecture	13	23.83	0.793	78.4
Classroom 203	Lecture	8	29.25	0.689	77.8
Chemistry 101	Lab	8	22.67	0.647	77.0
Chemistry 203	Lab	5	14.17	0.854	73.0
Classroom 205	Lecture	10	31.0	0.584	72.9
Classroom 170	Lecture	7	20.5	0.718	69.8
Chemistry 104	Lab	5	14.17	0.782	68.3
Chemistry 102	Lab	4	11.33	0.851	67.0
Chemistry 103	Lab	5	14.17	0.745	65.9
Classroom 204	Lecture	7	24.0	0.564	63.7
Chemistry 201	Lab	2	5.67	0.974	63.3

Table: Top 14 Rooms by Efficiency Score (Spring 2026)

3.2 Communication and Collaboration Approach

Consistent with the Utilization-Focused Evaluation model, our team prioritized stakeholder engagement throughout the evaluation process rather than reserving communication for a final report handoff. Our approach was structured around Chapter 14 and Chapter 16 of Russ-Eft and Preskill's framework for communicating and reporting evaluation activities.

Facilitated Working Sessions

We hosted two facilitated working sessions rather than presenting static slide decks. The first session was a smaller group consisting of our team, the VP of Facilities, and department chairs from the heaviest-use buildings. We presented our color-coded floor plan visualizations, which map each room by function (Lecture, Lab, Office, Research) and annotate efficiency scores for all scheduled rooms, allowing participants to see building-level patterns at a glance. The goal was not to present finished recommendations but to surface questions: Why does the Classroom Building carry 229 weekly scheduled hours while Olson Hall carries 74? Is the 0.63 average seat-fill a sign of scheduling inefficiency or pedagogical choice? Department chairs hold that institutional knowledge.

The second session was broader, including Registrar and Physical Plant stakeholders, and shifted from data interpretation to collaborative action planning. Peak-period heatmaps were shared and department chairs were asked what scheduling constraints drive the Monday-Thursday clustering and what would need to change to distribute load more evenly.

Reframing Negative Findings

Following the textbook's guidance on negative findings, we positioned low-utilization rooms as opportunities rather than deficiencies. Instead of stating "Room 106 is underutilized," we asked stakeholders, "Given this room's scheduling pattern, what alternative uses could support the medical school expansion?" This reframing invited creative thinking and avoided defensiveness, treating stakeholders as problem-solvers rather than as the source of the problem.

Reporting Plan

Audience	Format	Timing	Purpose
VP of Facilities	Both working sessions + ongoing updates	Feb-Apr 2026	Primary client; co-interpretation of data
Department Chairs	Working session 1 + draft review	Mar 2026	Verify data; contextualize findings
Registrar / Physical Plant	Working session 2	Mar 2026	Action planning and data verification
President Callahan	Executive briefing with floor plans	Apr 2026	High-level findings and recommendations
Board of Regents	Final report + floor plan presentation	Apr 2026	Decision support for capital planning

3.3 Stakeholder-Developed Action Plan

Through the facilitated working sessions, stakeholders collaboratively developed an action plan organized into three tiers:

Tier 1: Immediate Scheduling Optimizations (No Capital Investment)

Consolidate repeated low-fill rooms (Olson 103, 105; Classroom 126, 235) by redistributing their sections into higher-capacity rooms with available time blocks. Redistribute peak-period loads by shifting some Monday-Thursday midday sections to underutilized Friday and late-afternoon slots. Update EMS/Banner records to reflect verified room assignments and correct known data discrepancies identified through interviews.

Tier 2: Medium-Term Swing Space Strategies

If the Classroom Building goes offline for construction, Olson Hall's high-use rooms (120 and 114) should be preserved as priority swing space. Classroom 205, 203, and 104, which carry the largest instructional loads, require equivalent replacements (capacity, layout, technology) before any building can be decommissioned. Chemistry Building labs cannot be easily absorbed elsewhere without specialized swing space; temporary modular lab facilities may be required.

Tier 3: Long-Term Capital Planning

The evaluation data should be incorporated into the broader medical school feasibility analysis to determine which buildings are candidates for renovation versus replacement. The efficiency scores and floor plan visualizations provide a quantitative basis for the Board to evaluate the \$150M capital investment against demonstrated space needs. A follow-up evaluation should be conducted after any scheduling optimizations are implemented to measure the impact on the efficiency gap.

3.4 Evaluator Recommendations

Beyond the stakeholder-developed action plan, our evaluation team offers the following independent recommendations:

1. Institutionalize Data-Driven Space Management. This evaluation revealed that the university's current space allocation is largely anecdotal. We recommend the Registrar's Office and Physical Plant establish a recurring (semesterly) space utilization report using the efficiency score methodology developed in this evaluation. This would transition space management from a reactive to a proactive model.

2. Invest in System Modernization. The discrepancies between EMS/Banner data and actual room usage were significant and well-known among staff. We recommend the university invest in modernizing its scheduling infrastructure, potentially integrating real-time occupancy sensors, to reduce manual data entry errors and provide live utilization data.

3. Expand the Evaluation Scope. This study covered three buildings under a tight deadline. We recommend extending the same methodology to the full campus to build a comprehensive space inventory before committing to the medical school construction timeline.

4. Protect Lab Capacity. Chemistry Building labs operate at the highest utilization (0.76 avg seat-fill) and are the most expensive to replicate. Any relocation plan should prioritize securing specialized lab swing space before breaking ground on new construction. The consulting firm's feasibility study estimates \$85M for a new 100,000 sq ft building; part of that budget should be ring-fenced for temporary lab facilities during the transition.

5. Leverage Floor Plan Visualizations for Ongoing Decision Support. The color-coded floor plans developed for this evaluation provide an intuitive, at-a-glance view of space function and efficiency across all three buildings. We recommend the university adopt this visualization approach for ongoing facilities planning, making updated floor plan reports available to the VP of Facilities and Registrar's Office as a standing decision-support resource each semester.

Appendix A: Data Collection Tools

A.1 Evaluation Proposal Letter (Email to Registrar / Dean's Office)

At the request of President Callahan, Physical Plant is initiating a comprehensive space and utilization study of the Classroom Building, Chemistry Building, and Olson Hall. I was approached by the university's VP of Facilities to help gather this data. The goal of this study is to develop a complete, data-informed picture of how these spaces are used throughout the academic year to support future planning and resource allocation.

To ensure the accuracy and completeness of this study, our team will need to gather and cross-reference data from several sources:

Classroom and Lab Scheduling Data: From the Registrar's office, we requested detailed EMS classroom, academic, class, and hourly utilization reports listing all scheduled courses, sections, and meetings held in both the Classroom and Chemistry buildings. We were particularly interested in the specific room assignment for each class/lab section, the days and times each room was scheduled for instructional use, and the course enrollment for each section.

Instructional Type: To distinguish between classroom and laboratory utilization, we requested assistance in identifying which scheduled sections were designated as laboratory instruction versus lecture/discussion sections.

Faculty and Staff Office Assignment: From the Dean's office, we requested a current list of assigned staff and faculty offices within these two buildings, including room numbers and the primary occupant's name and title.

A.2 Email Invitation for Key Informant Interview

Subject: Space Utilization Study - Brief Meeting Request (Action Required)

Dear [Name],

At the request of President Callahan and the VP of Facilities, our team is conducting a comprehensive space and utilization study of the Classroom Building, Chemistry Building, and Olson Hall. The goal is to develop a data-informed picture of how these spaces are used to support future capital planning and the potential medical school initiative.

We are currently reviewing the EMS and Banner scheduling data, but we know that systems don't always tell the whole story. We would like to schedule a brief, 15-20 minute interview with you to discuss how [Department Name] actually utilizes its assigned spaces, verify office assignments, and identify any discrepancies in the current system.

Thank you in advance for your partnership. Your insights are crucial to ensuring the Board of Regents receives an accurate representation of your department's needs.

Sincerely,

Samuel C., Sam B., Veronica H.

Evaluation Project Team

A.3 Archival Data Extraction Parameters

Required data fields per scheduled section:

Field	Description
building_name / room_number	Building and room identifier
space_type	Lecture vs. Laboratory designation
meeting_days	Days of instruction (M, T, W, R, F)
start_time / end_time	Scheduled meeting times
weekly_hours	Total duration of meeting times per week
course_enrollment	Actual live roster count
room_capacity	Official maximum occupancy

This data was ingested into our analysis tool to calculate the avg_seat_fill and efficiency_score for each room using the formula: $\text{Efficiency} = 100 * (0.65 * \text{Utilization} + 0.35 * \text{Intensity})$.

Appendix B: Color-Coded Floor Plans

The following floor plans were developed by the evaluation team using verified room data. Rooms are color-coded by function: Lecture (blue), Lab (green), Office (yellow), and Research (orange). Efficiency scores are annotated for all scheduled rooms. These visualizations were presented to President Callahan and university leadership as part of the evaluation deliverables.

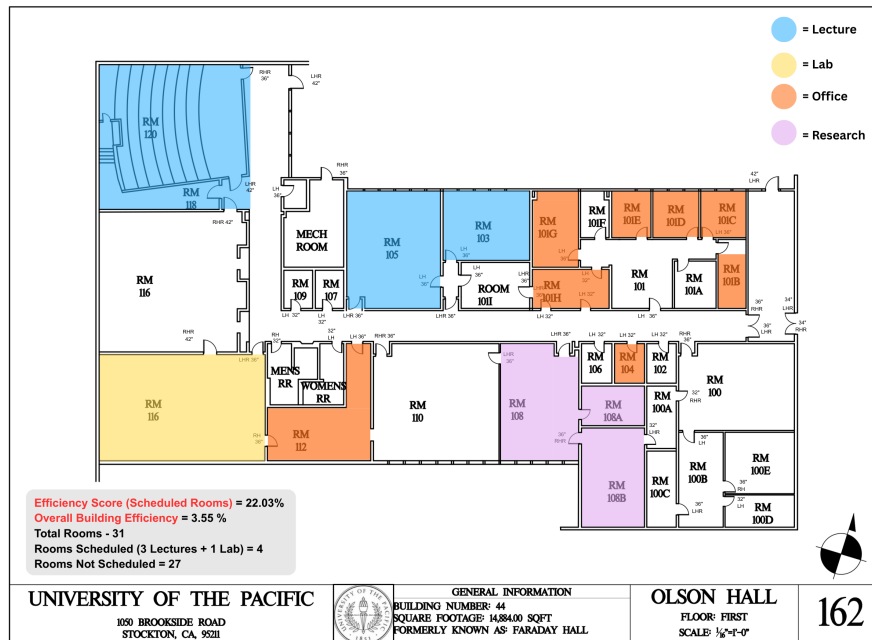


Figure: Classroom Building - First Floor

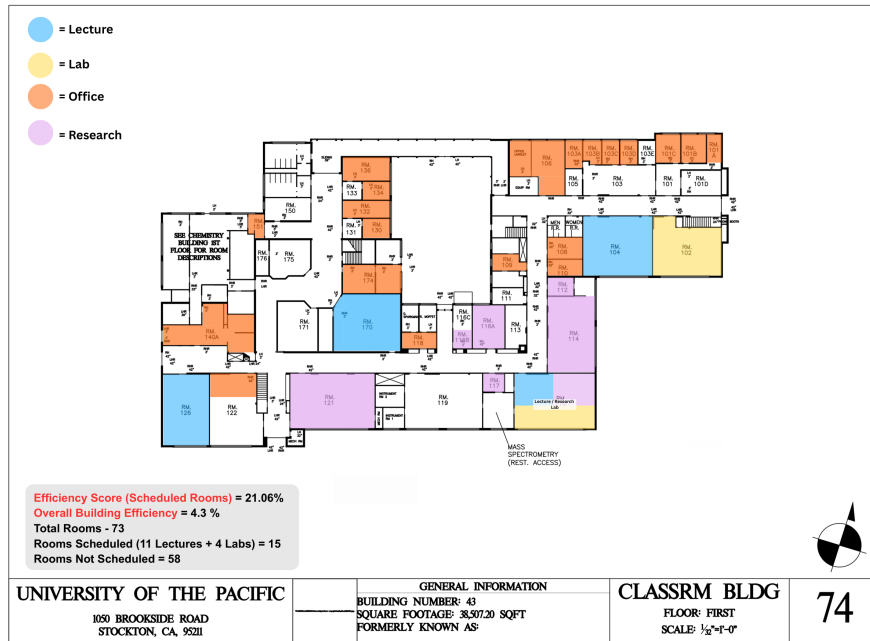
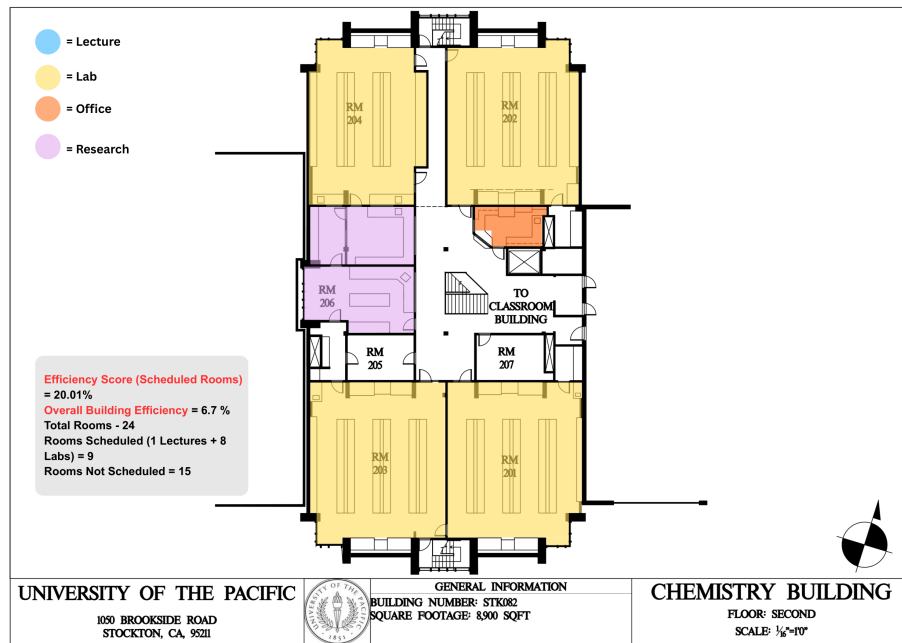


Figure: Classroom Building - Second Floor (with Legend and Efficiency Summary)



5

Figure: Chemistry Building - Detailed Floor Plan

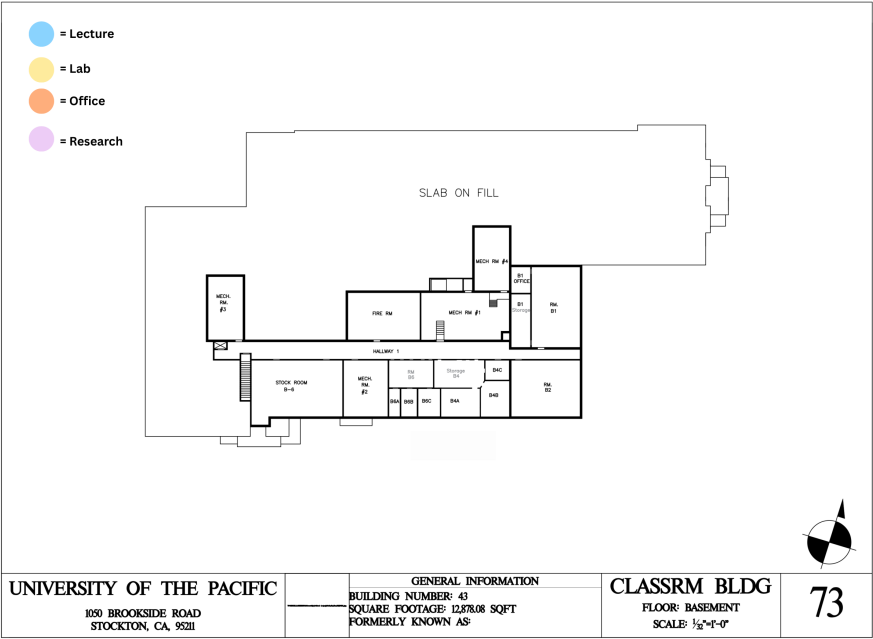


Figure: Olson Hall